

AIR TRANSPORTATION SYSTEM

VI Semester										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE24		PCC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I		AVIATION INDUSTRY								
Introduction, history of aviation-evolution, development, growth, challenges. Aerospace industry, air transportation industry-economic impact- types and causes. Airline Industrystructure and economic characteristics.The breadth of regulation- ICAO, IATA, national authorities (DGCA, FAA).Safety regulations-risk assessment-human factors and safety, security regulations.										
UNIT-II		AIRSPACE CATEGORIES								
Categories of airspace-separation minima, airspace sectors-capacity, demand and delay. Evolution of air traffic control system-procedural ATC system, procedural ATC with radar assistance, first generation 'automated' ATC system, current generation radar and computer-based ATC systems. Aerodrome air traffic control equipment and operation-ICAO future air-navigation system service provides as businesses.										
UNIT-III		AIRCRAFT OPERATIONS								
Aircraft: Costs-project cash-flow, aircraft price. Compatibility with the operational infrastructure.Direct and indirect operating costs. Balancing efficiency and effectiveness-payload-range, fuel efficiency, technical contribution to performance, operating speed and altitude, aircraft field length performance, typical operating costs. Effectivenesswake-vortices, cabin dimensions, flight deck.										
UNIT-IV		AIRPORTS FOR OPERATIONS								
Airports set up-airport demand, airport runway characteristics-length, declared distances, aerodrome areas, obstacle safeguarding. Runway capacity- evaluating runway capacity –sustainable runway capacity. Runway pavement length, Maneuvering area- airfield lighting, aprons, Passenger terminals-terminal sizing and configuration. Airport demand, capacity and delay.										
UNIT-V		AIRLINES								
Introduction -Setting up an airline-modern airline objectives. Route selection and development, airline objectives.Route selection and development, airline fleet planning, annual utilization and aircraft size, seating arrangements.Indirect operating costs. Aircraft- buy or lease. Revenue generation, Computerized reservation systems, yield management. Integrating service quality into the revenue-generation process.Marketing the seats. Airline scheduling-financial viability, regularity compliance.										
Text Books:										
1.Bazargan,M., 'AirlineOperationsandScheduling', 2 nd edn.,AshgatePublishingLtd,2010										
Reference Books:										
1. Belobaba,P.,Odoni,A.,Barnhart,C. 'TheGlobalAirlineIndustry',Wiley,2009. 2.Cheng-Lung, 'AirlineOperationsandDelayManagement',AshgatePublishingLtd,2010. 3. Wensveen,J.G., 'AirTransportation:AManagementPerspective', 6 th edn.,AshgatePublishingLtd,2007. 4.Yu,G., "OperationsResearchinAirlinesIndustry".AcademicPublishers,1998										

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Understanding the Foundations of Aviation and safety regulations in aviation and their importance.
2. Define and differentiate the various categories of airspace. Identify and analyze different airspace sectors. Evaluate the benefits and challenges associated with the current generation of ATC technology.
3. Evaluate the compatibility of aircraft with existing operational infrastructure. Analyze the trade-offs between payload, range, and fuel efficiency. Evaluate the impact of technical advancements on operational efficiency.
4. Understand the requirements for runway pavement length based on aircraft types and operational considerations.
5. Evaluate the considerations in determining seating arrangements on aircraft. Evaluate the financial viability of airline schedules.